Gregory Brooks

Senior Firmware Engineer at SuperSharp Space Systems Ltd.

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Relevant Experience

2025 to present — **Senior Firmware Engineer**, *SuperSharp Space Systems Ltd.*, Cambridgeshire.
Senior firmware engineer, developing flight firmware for real-time mechatronic control of satellite deployment mechanisms.

2022 to 2025 — **Medical Software Consultant**, *Team Consulting Limited*, Cambridgeshire.
- Software lead for the iF Design Award winning NANOme system (https://ifdesign.com/en/winner-ranking/project/nanome/642972).
- IEC 62304 compliant software development for medical devices; I've worked as a project software lead at every level of the V-model from early stage system/software architecture to detailed design and implementation, to testing and release.
- Performed early-stage project feasibility assessments and development proposals.
- Applied continuous integration and delivery techniques and technologies to facilitate rapid iterative development cycles and automated testing (containerised build environments running in Bitbucket Pipelines).
- Designed message-passing APIs using protobuf for asynchronous stream based communication between applications.
- Developed software for bare-metal, RTOS (FreeRTOS) and Linux (Ubuntu) environments.
- Exploratory work with Rust and Go.
- Sat on the 2024 Winter Party planning committee.

2021 to 2022 — **Senior Embedded Software Engineer**, *Signaloid Limited*, Cambridge.
Worked with a wide range of technologies, including:
- AWS IoT services and SDK for streaming sensor data from a BLE enabled IoT device to Signaloid's cloud infrastructure via MQTT.
- High level hardware system architecture for FPGA based compute modules.
- Software development in C and C++ for Signaloid's uncertainty-tracking processor (see github.com/signaloid for examples).

2019 to 2021 — **Consultant (Embedded Systems)**, *TTP plc*, Melbourn.
Worked on a range of multidisciplinary projects within the Life Science/Cell and Gene team, such as:
- CoVent ventilator, a collaboration with Dyson in response to the 2020 'Ventilator Challenge' during the COVID-19 pandemic.
- Puckdx sample-to-answer human IVD platform (for DiaSorin).

Responsibilities included:
- Hardware (schematic and PCB) design, assembly of prototypes.
- Firmware development for STM32 family, from bare-metal C99 to mutithreaded C++11 with mBed RTOS.
- Software/GUI development in Python (Kivy touchscreen interface for an embedded SBC).
- Testing and debugging of hardware, firmware and software (oscilloscopes, logic analysers, SWD debuggers).
- Implementation, testing and debugging of communications interfaces (RS485, I2C, SPI, Modbus, Ethernet<->TCP<->HTTP).
- Communication and collaboration with other team members, especially those without an electronics/software background (e.g., scientists, project managers).
- Project timeline estimation, prioritisation of tasks to meet aggressive deadlines.

Education

2015–2019 **MEng & BA Electrical and Information Sciences**, *Christ's College, University of Cambridge*, *2.1*.

2008–2015 **A-level Mathematics, Futher Mathematics, Physics & Chemistry**, *Sutton Grammar School*, Sutton, *4 A* grades*.

Activities & Awards

- Co-authored and presented a poster at the EuroSys 2019 conference in Dresden *(Gregory Brooks, Youchao Wang and Phillip Stanley-Marbell. Safeguarding Sensor Device Drivers Using Physical Constraints. Poster presented at EuroSys 2019, Dresden, Germany.)*.
- Won Google Creative Technology Prize at the national Big Bang Science Fair 2015.
- At university, I was a core member of Cambridge University Spaceflight society where I have worked on the design, construction, programming and testing/flight of projects such as an inertial measurement unit, GPS/telemetry boards for rockets, lightweight balloon payloads and a time-of-flight trilateration system for tracking a rocket's position during flight.
  - Developed Python applications and GUIs using PyQt for various society projects e.g., the trilateration project mentioned above.
  - Part of the team that launched the society's Martlet 3 rocket at Black Rock desert, Nevada, in 2017.

Master's Degree Project

Title *Compiling Physical Invariant Descriptions to Hardware Descriptions for a Sensor Interface for Security and Privacy in IoT Applications*

Supervisor Dr Phillip Stanley-Marbell

Description This project involves writing a compiler, in C, that takes a description of physical laws/constraints relating electronic sensor data (e.g., pressure $\times$ volume $\propto$ temperature) and outputs Verilog RTL for use with a low power iCE40 FPGA. This FPGA sits between the sensors and external circuitry, such as a microprocessor, implementing a local differential privacy system which accounts for the physical relationships and hence mutual information between related measurements.

Degree Modules (Condensed Summary)

- Mathematics
- Signal Processing
- Information Theory
- Embedded Systems
- Analogue and Digital Electronics
- Software Engineering